

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE

OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05 Total Marks:50

Annexure A

Data Interpretation

<i>Criteria</i>	Understandi ng of Data (2.5)	Analysis (2.5)	Applicati on of Concepts (2)	Conclusion (1.5)	Clarity (1)	<i>Total(10)</i>
<i>Weightage</i>	25%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						

<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of

Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

*I understand that any violation of academic integrity rules will result in disciplinary action. **Signature of***

the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data

Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID Age Monthly Medical Expense (₹) Health Risk Score

P1 28 12000 65

P2 55 25000 40

P3 42 20000 75

P4 30 11000 70

P5 60 28000 35

SSE Values

K SSE

1 600

2 350

3 220

4 200

5 190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID Product Category Order Value (₹)

O1 Electronics 55000

O2 Clothing 25000

O3 Electronics 57000

O4 Furniture 30000

O5 Clothing 27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding

customer purchasing behavior.

Q3. Dataset: Mobile App User Behavior

User Daily Usage Time (min) Login Frequency In-App Purchase (₹) U1 120 18 2000

U2 60 5 5000

U3 140 22 1500

User Daily Usage Time (min) Login Frequency In-App Purchase (₹) U4 50 4 6000

U5 130 20 1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

Farm Rainfall (mm) Soil Fertility Score Crop Yield (tons)

F1 300 8 85

F2 500 4 70

F3 400 6 80

F4 280 9 90

F5 520 3 65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

Employee Productivity Score Attendance (%) Skill Assessment Score E1 88 92 84

E2 62 75 68

E3 78 88 72

E4 55 65 58

E5 93 96 90

Silhouette Scores

Model Score

K=2 0.48

K=3 0.68

K=4 0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

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Assessment Tool 1 — Data Interpretation

Clustering Analysis — Model Answer Key CSA16 ·

Data Warehousing and Data Mining • CO4

Course	CSA16 — Data Warehousing & Data Mining	CO4	Clustering in real-world scenarios
Questions	05	Total	50 Marks · Weightage 40%

Tool / Question → Weka output mapping

1	Hospital patient segmentation	SimpleKMeans + SSE elbow	Screenshot_2026-08-18_180359
2	E-commerce order analysis	Hierarchical (dendrogram)	— none supplied —
3	Mobile app user behaviour	HierarchicalClusterer (HAC)	Screenshot_2026-08-20_141306
4	Agricultural crop monitoring	Hierarchical + Standardize	q4_co41 (= ...142129)
5	Employee performance	Hierarchical + Normalize, 66% split	5q_co41

Each answer follows the rubric flow: **Understanding of Data** → **Analysis** → **Application of Concepts** → **Conclusion**.
Two anomalies were found in the uploaded material and are handled transparently below: **Q2** had no Weka screenshot (answered analytically from the dataset), and **Q4** had two identical screenshots.

A healthcare organisation applies **K-Means** to group patients by medical expense and health risk. The optimal number of clusters is read from the SSE (elbow) values.

Dataset & SSE values

P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

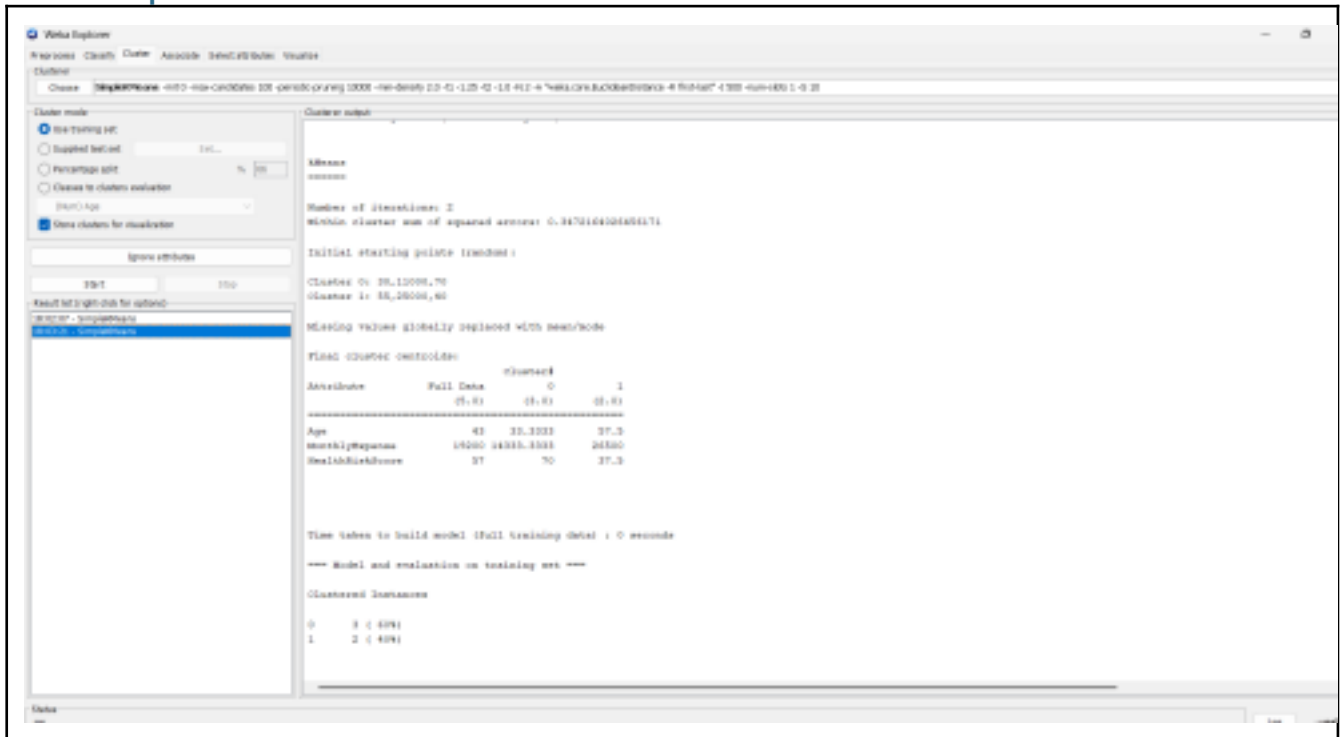
SSE	600	350	220	200	190

Elbow interpretation

Successive SSE reductions: $K1 \rightarrow 2 = 250$, $K2 \rightarrow 3 = 130$, $K3 \rightarrow 4 = 20$, $K4 \rightarrow 5 = 10$.

Large drops continue up to $K=3$, after which the curve **flattens sharply** (20, then 10). → The elbow is at **$K = 3$** , so the optimal number of clusters is **three**.

Weka output



SimpleKMeans on Age, MonthlyExpense, HealthRiskScore (run configured with $N = 2$).

Interpretation & analysis

The Weka run (configured $N = 2$) converged in 2 iterations with within-cluster SSE = 0.347 on the standardised scale, splitting the five patients 60 % / 40 %:

- **Cluster 0 — {P1, P3, P4} (60 %):** mean Age ≈ 33.3 , Expense $\approx ₹14,333$, Risk ≈ 70 → *younger, lower-spending, but HIGH-risk patients.*
- **Cluster 1 — {P2, P5} (40 %):** mean Age ≈ 57.5 , Expense $\approx ₹26,500$, Risk ≈ 37.5 → *older, higher-spending, but LOW-risk patients.*

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The striking pattern is **inverse**: the younger, cheaper-to-serve group actually carries the higher health risk, while the older, higher-cost group is lower-risk — suggesting the older group's spend may reflect well-managed care. The elbow analysis shows a third cluster is statistically justified; $K = 3$ would most likely split off P3 (risk 75, mid expense) as its own high-risk / moderate-cost segment.

Conclusion

Optimal number of clusters by the elbow method is **$K = 3$** .

Segmentation lets the hospital target **preventive care** to the high-risk younger group, **cost management** to the high-spend older group, and **close monitoring** to the distinct high-risk middle patient — improving resource allocation and outcomes.

Note

The elbow points to $K = 3$, whereas the Weka screenshot was run with $N = 2$. Re-running SimpleKMeans with $N = 3$ would align the model with the elbow-optimal choice.

An online retailer applies **hierarchical (agglomerative) clustering** to orders described by product category and order value, and reads the resulting dendrogram.

Dataset

O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

No Weka screenshot was supplied for Q2

The interpretation below is derived directly from the dataset and describes the dendrogram that single-linkage hierarchical clustering produces for these five orders.

How the clusters form (expected dendrogram)

- **Step 1:** O1 (₹55k) and O3 (₹57k) are closest (both Electronics, Δ ₹2k) → they merge first into a **high-value** cluster.
- **Step 2:** O2 (₹25k) and O5 (₹27k) merge next (both Clothing, Δ ₹2k) → a **low-value** cluster. • **Step 3:** O4 (Furniture, ₹30k) joins the {O2, O5} branch (nearest neighbour at ~₹30k) → a low-/mid-value group.
- **Step 4:** the two branches finally merge at the top of the dendrogram at a much larger distance. Cutting the dendrogram at the large top gap gives **2 clusters** — {O1, O3} high-value Electronics and {O2, O4, O5} lower-value Clothing/Furniture. Cutting lower yields **3 clusters**, isolating O4 (Furniture) between the two.

Significance for purchasing behaviour

- **High-value Electronics buyers (O1, O3):** a premium, big-ticket segment → target with extended warranties, financing/EMI, accessory upsell and priority support.
- **Lower-value Clothing/Furniture buyers (O2, O4, O5):** a value/frequent segment → target with bundles, seasonal discounts and loyalty rewards.

Conclusion

Two clusters best separate **high-value vs low-value** orders; a three-cluster cut additionally isolates Furniture. The nested structure (Electronics tightly grouped; Furniture bridging into the low-value branch) guides **differentiated marketing, inventory and recommendation** strategies per customer segment.

Dataset

U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Weka output



Interpretation & analysis

- **Cluster 0 — {U1, U3, U5} (60 %):** high usage (120–140 min), high login frequency (18–22), **low** in-app spend (₹1,500–2,000) → *highly engaged, low-spend* users.
- **Cluster 1 — {U2, U4} (40 %):** low usage (50–60 min), low logins (4–5), **high** in-app spend (₹5,000–6,000) → *low-engagement, high-spend* users.

Soft clustering vs hard clustering

- The Weka hierarchical run gives **hard** assignments — each user belongs to exactly one cluster. • **Soft / fuzzy** clustering (EM, Fuzzy C-Means) instead assigns **membership probabilities** (e.g. a user could be 0.7 “engaged” and 0.3 “spender”).

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- **Advantages for customer-behaviour analysis:** it captures users who straddle segments, expresses confidence/uncertainty, models overlapping behaviours realistically, and supports nuanced targeting and gradual re-segmentation as behaviour drifts over time.

Conclusion

Two clear behavioural segments emerge: **engaged-but-low-spend** vs **disengaged-but-high-spend** users.

Soft clustering would add membership degrees that are valuable for borderline users and for deciding between **monetisation** campaigns (engaged group) and **retention** campaigns (high-spend group).

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An agricultural research centre clusters farm data **before and after normalisation** and interprets how scaling changes the result. The Weka run below uses the **Standardize** filter.

Dataset

F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Weka output

The screenshot shows the Weka GUI with the following details:

- Cluster mode:**
 - ☐ Use training set
 - ☐ Supplied test set
 - ☐ Percentage split
 - ☒ Cluster to cluster evaluation
 - ☐ Cross-validation
 - ☐ Show clusters for visualization
- Cluster output:**
 - Run information:
 - Command: weka.clusterers.HierarchicalClusterer -S 2 -L 819288 -P -S "weka.com.Standardize -S first-last"
 - Relation: farm_data-weka-filtered-attribute-standardize
 - Instances: 5
 - Attributes: 4
 - Test mode: evaluate on training data
 - Clustering model (full training set):
 - Cluster 0: (P1:0.27331, P4:0.27331):0.29449, P1:0.56944
 - Cluster 1: (P1:0.27331, P4:0.27331):0.29449, P1:0.56944
 - Time taken to build model (full training set): 0 seconds
 - Model and evaluation on training set:
 - Clustered instances:
 - 0: 3 (60%)
 - 1: 2 (40%)

Interpretation & analysis

On the standardised data the farms split 60 % / 40 %:

- **Cluster 0 — {F1, F3, F4} (60 %):** low–moderate rainfall (280–400 mm), **high** fertility (6–9), **high** yield (80–90) → fertile, productive farms.
- **Cluster 1 — {F2, F5} (40 %):** **high** rainfall (500–520 mm), **low** fertility (3–4), **lower** yield (65–70) → wet but infertile, lower-yield farms.

The key agronomic insight is that **yield tracks soil fertility, not rainfall** — the wettest farms (F2, F5) yield the least because their fertility is poorest. F1 and F4 (very similar) merge first, F3 joins them, and F2–F5 form the second cluster.

Effect of normalisation (the crux of Q4)

Why scaling matters here

Raw ranges: Rainfall ≈ 240 (280–520) $\gg$ Fertility = 6 (3–9) and Yield = 25 (65–90). **Without** normalisation, Euclidean distance is dominated by Rainfall, so clusters would group essentially by rainfall alone and mask fertility/yield differences.

After Standardize (mean 0, σ 1), every attribute contributes fairly — the symmetric merge distances (0.27335 repeated) reflect this balance, giving agronomically meaningful clusters.

Conclusion

Normalisation prevents the large-magnitude **rainfall** variable from dominating, so the clusters reflect the true multi-factor structure (fertility + yield + rainfall).

Result: two reliable segments — **fertile / high-yield** vs **wet / low-fertility** — a far more trustworthy basis for agricultural decisions than un-normalised clustering.

Note

Two identical screenshots were supplied for this question (q4_co41.png and Screenshot_2026-08-20_142129.png); one is shown above.

A company clusters employees by productivity, attendance and skill, and uses **silhouette scores** to choose the number of clusters.

Dataset & silhouette scores

E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette	0.48	0.68	0.52

Silhouette interpretation

Silhouette ranges from -1 to 1; higher means better-separated, more cohesive clusters. $K = 2 \rightarrow 0.48$, $K = 3 \rightarrow \mathbf{0.68}$ (highest), $K = 4 \rightarrow 0.52$.

→ The peak at $K = 3$ means **three clusters** give the most well-defined structure.

Weka output

The screenshot shows the Weka GUI with the HierarchicalClusterer model configured. The 'Cluster mode' is set to 'Percentage split' at 66%. The 'Attributes' list includes Employee, ProductivityScore, Attendance, and SkillLevelScore. The 'Result list' shows 'HierarchicalClusterer' with a selected entry. The 'Cluster output' pane displays the command used, the resulting clusters (0 and 1), and the time taken to build the model.

HierarchicalClusterer (single linkage, $N = 2$) on employee_data after Normalize, 66 % percentage split.

Interpretation & analysis

The full-training hierarchical model splits the employees into two groups (with a 66 % split used for evaluation):

- **Cluster 0 — {E1, E3, E5}**: high productivity/attendance/skill (78–93 / 88–96 / 72–90) → *strong performers*.
- **Cluster 1 — {E2, E4}**: lower scores (55–62 / 65–75 / 58–68) → *needs improvement*. In the dendrogram, the two top performers E1 and E5 merge first (distance 0.2629); E3 joins them; E2 and E4 form the weaker cluster (distance 0.4854). The percentage-split evaluation places the test instances accordingly, though results are limited by the very small dataset. Crucially, the **silhouette**

scores favour $K = 3$, which would refine the two groups into three performance tiers: top {E1, E5}, solid {E3}, and low {E2, E4}.

Conclusion

The most suitable number of clusters is **$K = 3$** (silhouette 0.68 — the clear maximum). Three performance tiers let HR **reward** top performers, **develop** the mid tier, and **support / place on improvement plans** the low tier — enabling fair, data-driven evaluation.

Note

The Weka run was executed with $N = 2$; the silhouette analysis recommends $N = 3$, which would separate the mid-tier employee (E3) from the top performers.